

Impact of motor error processing on performance on a working memory task: effect of modulating cognitive load in high and low span groups[☆]

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ABSTRACT

Making a motor error, hitting the wrong key for example in responding to a stimulus, impacts perceptual, motor, and cognitive processes by implying a variation in performance and response time on the next trial (Post Error Slowing, Dutilh et al., 2012; Wessel, 2018). Wessel et al. (2022) showed that an error produced on a flanker-type task had a negative impact on a working memory span recognition. This effect is called ERIAM (Error Related Impairment of Active Working Memory) and consists of degradation of memory span following the production of an error on a conflict task concurrent to a memory task. In their study, the ERIAM effect is more pronounced in Low Span individuals. This suggests that these individuals have less stable Working Memory representations and are more likely to be impacted by an error. The presented study replicates the ERIAM protocol of Wessel et al. (2022) by adding variation of cognitive load. In addition, span recognition task is modified with a recall task. Results show a replication of ERIAM effect in our sample. Moreover, ERIAM effect is only significant in lowspan group in the moderate cognitive load condition and in highspan group in the increased load condition. In line with Lavie's load theory (Lavie et al., 2004), we explain these results by the fact that High and Low Span groups do not process cognitive load increase in the same way.

1. Introduction

The investigation of post-error phenomena has been an area of scholarly inquiry since the 1960s (Danielmeier & Ullsperger, 2011; Dudschig & Jentsch, 2009; Dutilh et al., 2012; Laming, 1979; Notebaert et al., 2009; Pfister & Foerster, 2021), following the publication of Rabbitt's seminal article (Rabbitt, 1966). Since the field's inception, research has been predicated on the examination of behavioral phenomena that emerge in the aftermath of errors. These include the lengthening of response times (RTs) and the subsequent modification of accuracy. Such behavioral adjustments termed Post Error Adjustments (PEA) (Danielmeier & Ullsperger, 2011), serve as an indirect means of inferring underlying cognitive process. Thus, PEA have been linked to cognitive control and working memory implementation (Regev & Meiran, 2014). The presence of PEA has been documented in a variety of conflict tasks, including the flanker task (Debener et al., 2005), the Stroop task (Gehring & Fencsik, 2001), and the Simon task (King et al., 2010). In a dual-task procedure intertwined with a working memory

task, Wessel et al. (2022) collected data demonstrating the impact of a motor error on working memory maintenance. This study provided experimental evidence linking errors to higher-order cognitive functions including working memory, cognitive control and attention.

To examine the nature of errors, it is essential to consider the Error Generic Error Modelling System (GEMS), developed by Reason (1990). GEMS provides a robust framework for classifying errors based on the level of cognitive processing achieved. The system distinguishes three categories of error: skill-based, rule-based, and knowledge-based. This study focuses primarily on the first level, designated as 'skill-based' (see Skill-Rule-Knowledge (Rasmussen, 1983), from which Reason's model is derived). This level encompasses errors resulting from attentional lapses or to habitual responses. Such errors arise from a failure to attend to relevant aspects of the information being processed and may also stem from interference by irrelevant features of the task or stimulus. This phenomenon is particularly evident in conflict tasks, where attentional resources are diverted from task objectives. Cognitive systems have been shown to monitor these errors (Botvinick et al., 2001), and to support

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their detection and correction through the engagement of cognitive control mechanisms.

In line with the seminal work of Miyake et al. (2000), cognitive control is conceptualized as a triad of factors: the inhibition of a prepotent response, the updating of working memory, and the shifting between mental sets. These components exhibit distinct variance and correlations, as elucidated by Friedman and Robbins (2022). Cognitive control is immediate upon error detection (Shalgi et al., 2009), and serves to monitor such errors (Botvinick et al., 2004). As demonstrated by Unsworth et al. (2022), individuals with high working memory capacity display a reduced susceptibility to attentional fluctuations during task execution. These individuals exhibit an enhanced ability to oversee errors (i.e., error monitoring). Conversely, individuals with low working memory capacity have been found to show a greater tendency towards unperceived errors, phenomenon referred to as error blindness (Buzzell et al., 2017), which is attributable to attentional fluctuations or mind wandering (Christoff et al., 2016). Hence, cognitive control has been shown to play a critical role in managing perceived errors and is related to attention. Error detection requires individuals to maintain sufficiently alert state, which imposes a high demand on attentional resources. Error awareness has been found to correlate negatively with attentional deficits and positively with working memory capacities (WMC). As proposed by Botvinick et al. (2001), the cognitive system evaluates the necessity for cognitive control by detecting conflicts in information processing. Upon the occurrence of an error, this system is engaged (Botvinick et al., 2004) to resolve the conflict arising from the discrepancy between the executed response and the expected response. Once the conflict is resolved, the individual can reorient their focus to the task at hand, allocating cognitive and attentional resources more effectively. Error monitoring and the behavioral adjustments that occur in response to errors have been shown to offer valuable insights into underlying attentional and cognitive processes (Botvinick et al., 2004). It has been shown to be indicative of attentional lapses (Unsworth et al., 2022) and cognitive control processes (Regev & Meiran, 2014). According to Engle and Kane (2003), a strong correlation exists between WMC, attentional control, and the conflict management system. Improved WMC has been shown to facilitate enhanced conflict detection and more efficient implementation of attentional control mechanisms (Unsworth et al., 2022). Consequently, individuals with superior WMC (and, by extension, enhanced attentional control) are likely to exhibit greater efficacy in error processing (Unsworth et al., 2022).

“Everyone knows what attention is. It is the taking possession by the mind, in clear and vivid form, of one out of what seem several simultaneously possible objects or trains of thought.” (James, 1890). Attention plays a crucial role in information processing. As proposed by the models of Egeth and Kahneman (1975) and Wickens (2008), the limited capacity of human information processing capabilities is constraint by the finite attentional resources available. Consequently, the selection of relevant information is a prerequisite for achieving an individual's goal (Niedermeyer, 1994). Selectivity is defined by the differentiation between two distinct functions of attention: the first being divided attention, which conceptualizes attention as a reservoir of processing resources; and the second being selective attention, which characterizes attention as an instance of conscious control of processing. The hypothesis that perception is capacity-limited was first introduced at the perceptual level (Broadbent, 1958; Treisman, 1964). Subsequently, the late selection hypothesis was proposed by Deutsch and Deutsch (1963), suggesting that stimulus perception operates as an automatic process, bringing all stimuli to be processed in depth by the cognitive system.

In order to achieve the selection of information at this advanced level of cognitive processing, attentional control also has an important role to play. Attentional control refers to the ability to focus and sustain attentional resources on a specific task or goal, while filtering out irrelevant information, whether from internal thoughts or external stimuli. When attentional control is required, attentional resources are endogenously oriented (i.e., top-down), in accordance with an

individual's goals and objectives. These resources are allocated in a controlled and voluntary manner to relevant aspects of the environment and/or internal cognitive processes to facilitate goal achievement. The prefrontal cortex (particularly the dorsolateral area) supports the active maintenance of task-related goals and has been associated to the individuals' WMC (Kane & Engle, 2002). To gain insight into the determinant of the process of selecting relevant information from perceptual to cognitive levels, Lavie introduced a theory that measures the impact of perceptual and cognitive loads.

Indeed, Lavie and Tsai (1994) proposed the perceptual load theory (see also Lavie and Dalton (2014) and Lavie (2010)), which models selective attention and resolves the debate about the location of attentional filtering. According to this theory, the primary determinant of the locus of attentional filtering is perceptual load. In order to test this, Forster and Lavie (2007) proposed a visual search task in which participants were required to detect the presence of a target letter—predefined as either ‘X’ or ‘N’—within a circle composed of items. A distractor letter was simultaneously displayed adjacent to the circle. This distractor letter either matched the target letter constituting the ‘congruent condition’, e.g. target ‘X’ and distractor ‘X’ or other alternative (the ‘incongruent condition’, e.g. target ‘X’ and distractor ‘N’). The impact of the distractor was quantified by measuring differences in response time between congruent and incongruent conditions. In incongruent trials, response times were observed to increase. In the low-load condition, the circle consisted of small, homogeneous, non-target elements (the letter ‘o’) thereby increasing the visual salience of the target letter (Posner, 1980). Conversely, in the high load condition, the circle was composed of multiple letters including the target letter. The authors observed that, under high perceptual load, the distractor's effect on response time was almost entirely cancelled. This suggests that increased perceptual load reduces or even cancel the distractor's effect, and in this particular instance, an early attentional filtering process limits the perception of irrelevant elements.

When the perceptual load is minimal, individuals' residual processing abilities may be influenced by irrelevant distractor stimuli. This suggests the necessity for subsequent selection processes to be implemented in order to prevent these distractors from exerting control over behavior (Lavie, 2010). Lavie and Dalton (2014) introduced a dual-task protocol to measure the effect of cognitive load on distractor processing. The protocol involved the administration of a memory task, in which participants were required to memorize a digit span, followed by a selective attention task, prior to the recognition of the digits presented first. In the selective attention task, participants were instructed to identify a centrally presented target letter (‘z’ or ‘x’), which appeared concurrently with a peripheral distractor letter. The distractor was either congruent, incongruent, or neutral with respect to the target. This task constitutes a minimal perceptual load, as it involves only the target and the distractor letters. In the digit span task, cognitive load is manipulated by varying the number of digits presented (from 1 to 6) in order to examine its effect on distractor processing. It is hypothesized that a higher cognitive load will increase the distractor's effect on response time, which is the opposite effect to that of a higher perceptual load.

When stimulus presentation engenders a high level of perceptual load, attentional filtering is early and automatic, leading to the saturation of the individual's capacity. Distractors that are not relevant to the task at hand may not even be perceived. However, when this level allows for all perceived information (both relevant and irrelevant) to be processed without overload, filtering will occur at the higher (cognitive) level of processing. The consequences of this filtering process depend on its location. In cases where attention filtering occurs at an early stage, distractibility to irrelevant stimuli is minimized, as such stimuli are filtered out and, therefore, not perceived. Conversely, low perceptual load has been shown to result in an increased susceptibility to irrelevant stimuli. Furthermore, higher cognitive load has been demonstrated to exacerbate distractors' impact. The effects of perceptual and cognitive

load on the processing of distractors, such as task-irrelevant stimuli, are contradictory. Therefore, present study will focus on the cognitive load effect on attentional processes.

The observed differences in WMC between individuals can be attributed to variations in attentional control. To investigate the strategies employed by diverse participant groups in the selecting of relevant information while filtering out irrelevant content, Kane and Engle (2000) assessed the performance of two participant cohorts differentiated by their WMC (high span vs. low span). In their experiment, participants were presented a list of words. During the encoding phase, they were required to tap a simple sequence on a keyboard in time with a predetermined rhythm. This was followed by an interfering task, an oral version of the Trail Making Test (Armitage, 1946), lasting 16 s, designed to prevent rehearsal of the word list. Participants were subsequently asked to recall the list. Initial trials revealed comparable performance between high and low span participants. However, in later trials high span participants outperformed their low span counterparts. This phenomenon can be attributed to the proactive interference inhibition exhibited by high span individuals. Their superior attentional control capacity enables them to effectively counteract the effects of interference, thereby ensuring enhanced recall efficacy.

The authors then introduced an increase in the complexity of the finger-tapping sequence performed by participants following the encoding of the word list. This results in an overall cognitive cost, thus making the procedure closer to a dual task. In this case, a greater effect of proactive interference in low span versus high span was no longer observed. The authors concluded that high span participants are no longer able to counteract the effects of proactive interference in situations of higher cognitive load. Indeed, attentional control is a voluntary and resource-intensive process. As cognitive load increases, conflict resolution (proactive interference) is impaired for high span participants since they are no longer able to deploy sufficient attentional resources. This has a subsequent impact on the word list recall. In the low span condition, attentional control is not used to counteract the proactive interference effects, resulting in a stronger effect of this interference (compared in high span) and no difference between load conditions.

In order to assess behavioral responses to errors, conflict tasks (e.g. flanker task, Simon task, Stroop task) are frequently employed in laboratory settings (Derrfuss et al., 2021; Schroder et al., 2020; Steinhäuser et al., 2017). These tasks are also commonly employed to demonstrate attentional control implementation (von Bastian et al., 2020). The design of these tasks requires participants to select one option from a series of alternatives as promptly as possible. The complexity of these tasks stems from the strategic manipulation of the relevant and irrelevant dimensions of the presented stimuli. Congruence is characterized by the similarity between the information provided by the target (relevant) and the distractors (irrelevant). Conversely, incongruence is defined by the divergence between these two dimensions. In congruent trials, participants typically exhibit a higher level of accuracy and faster response times compared to incongruent trials (Egner, 2007). The incongruent condition involves the interference of information from two distinct dimensions, resulting in response conflict and the subsequent activation of the conflict monitoring system. Incongruent trials have been shown to be more error-prone (Gratton et al., 2017). As a result, conflict tasks have been shown to induce errors for participants.

In the domain of error behavioral adjustments, Post-Error Slowing (PES) is defined as an increase in reaction time (RT) on post-error trials (Laming, 1979). Conversely, Post-Improvement of Accuracy (PIA) is characterized by an improvement in response accuracy on post-error trials. The relationship between response time and response accuracy on post-error trials is associated with the emergence of two predominant theoretical frameworks: The first, referred to as functional theories of PES (Fievez et al., 2022), is based on observations of increased accuracy levels in response to post-error trials. Conversely, the second framework, referred to as non-functional theories of PES, posit a decrease of accuracy in post-error trials. PIA is not consistently supported in the existing

scientific literature. The current body of research remains inconclusive, with some studies reporting a relationship between PES and PIA (e.g., Maier et al., 2011; Marco-Pallarés et al., 2008) while others do not (e.g., Carp & Compton, 2009; Fiehler et al., 2005). These theoretical accounts are also categorized as adaptive (functional) and maladaptive (non-functional). The improvement in post-error accuracy has been demonstrated to support the functional/adaptive explanations for PES. The body of adaptive theories provides a functional value to PES. In conjunction with the concomitant observation of increased accuracy on trials following the error (PIA: Danielmeier et al., 2011), PES enables participants to regain an optimal level of task performance. The association between subsequent increase in accuracy and PES has been modeled using a drift diffusion-type computational framework (Ratcliff & McKoon, 2008). This model shows a post-error increase in response caution materialized by a reduction in drift rate (Dutilh et al., 2011). The drift rate is model parameter that represents the speed at which information is accumulated from the stimulus. According to Dutilh et al. (2012), “[...] participants become more cautious after an error and slightly more daring after a correct response [...]”.

Conversely, in cases where post-error accuracy does not improve (Schroder et al., 2020), non-functional or maladaptive theories provide a more appropriate account for PES. These theories propose that PES arise from a diversion of attention towards task-irrelevant. An error, perceived as an unexpected event by the participant, thereby directs their attention towards it. However, when task difficulty increases and errors begin to outnumber correct responses, a phenomenon referred to as the post correct slowing effect is observed (Notebaert et al., 2009). This effect is characterized by an increase in reaction times (RTs) following a correct response, which is the opposite of the PES. In this framework, it's the unexpected aspect that causes the slowdown, and depending on the relative frequency of the stimuli, it can be either an error trial or a correct trial. In their protocol, errors are not followed by an improvement of subsequent performance (PIA), suggesting that PES has therefore no functional value. Other account within the maladaptive framework postulates an impossibility to process the additional information constituted by error. These theories posit that the processing of additional information, such as errors, is hindered by a bottleneck in the information processing capacity, resulting in a saturation of ongoing cognitive processes. By manipulating the response/stimulus interval, Jentzsch and Dudschig (2009) observed that the shortest intervals (50 ms) resulted in the highest PES, which was associated with a decline in response accuracy. At the longest intervals (1000 ms), PES was reduced relative to the shortest interval, and no significant change in response accuracy was reported.

Subsequent research has facilitated the integration of divergent data patterns from both adaptive and maladaptive theories accounts. Wessel (2017) synthesizes these contrasted approaches and highlights their complementarity through the Adaptive Orienting Theory (AOT). This theory postulates that PES unfolds in two distinct phases. The initial phase involves a broad inhibition of perceptual, motor and cognitive processes engaged to carry out the task. Attention is diverted from task-related goals due to the orienting response triggered by the unexpected nature of an error. However, this phenomenon is not specific to errors, as any unexpected event can elicit a similar pattern. This leads to a decrease in accuracy and an increase in RTs. Wessel refers to this as ‘orienting-related’ PES, which occurs when the subsequent stimulus is presented too early, during the ongoing inhibition period. Consequently, the occurrence of PIA is not observed, thereby facilitating the findings typically associated with non-functional (maladaptive) theories.

In the subsequent stage, Wessel (2017) introduces an error-specific phase in which the source of attentional disturbance is identified and examined. During this phase, attentional processes are progressively reallocated to the processing of errors. This is subsequently followed by a reengagement towards the objectives of the task and a recovery of pre-error performance levels. The purpose of this subsequent phase is to facilitate the restoration of optimal levels of accuracy in the task at hand.

Box 1

McIsaac et al. (2015) proposes a dual-task paradigms' taxonomy. A dual task is the concurrent performance of two tasks that can be performed separately, measured separately, and have distinct goals. Thus, each task may be different in character, drawing on distinct domains. For example, a memorization task will be based on mnemonic and cognitive processes. Another task may consist of performing a movement. The two tasks may also be based partly or totally on similar processes (e.g., two tasks of a cognitive nature). In Wessel et al. (2022), the experiment implemented involves a memory task and a motor task of deciding the direction of an arrow. The motor task requires selective attentional abilities, as it involves not being disturbed by distracting arrows of a flanker-type task (Eriksen & Eriksen, 1974). The memory task proposed by Wessel et al. (2022), taken independently, is a simple short-term memory task. Participants are asked to recognize a letter span presented a few seconds earlier. It is the performance of this task in a dual-task context that makes it a Working Memory task.

PES is thus described as 'strategic', a deliberate and gradual process influenced by cognitive control. Once these two phases have been completed, PIA can be observed, as postulated by adaptive theories. Therefore, PES has a functional status by allowing the refocusing of attentional processes on the current task to recover pre-error performance levels.

In their seminal work, Wessel et al. (2022) sought to establish a relation between motor error, working memory (WM) processes, and attentional processes. To this end, they proposed an experiment to test one of the predictions of Adaptive Orienting Theory (Wessel, 2017). This experiment sought to ascertain the impact of a motor error on ongoing cognitive processes. To this end, they used a dual-task protocol (see Box 1), comprising a memory task and a flanker task (Eriksen & Eriksen, 1974). The trial sequence is outlined as follows: Initially, a sequence of letters is displayed on the screen for a duration of 2000 milliseconds, requiring memory retention. Participants are then required to indicate the direction of a central arrow (the target, indicating left or right) while ignoring two distractor arrows symmetrically arranged to the left and right of this target. The distractor arrows both point in the same direction as the target (for the congruent trial) or in the opposite direction (for the incongruent trial). Following the completion of six trials of the flanker task, a sequence of letters is again presented to the participants. In 50 % of cases, the sequence is the same as the initially presented sequence and different in the other cases. Participants are tasked to evaluate the congruence between the two sets of letters through a YES/NO response. Prior to the initiation of the protocol, a training phase is employed to calibrate the size of the sequence of letters presented to the participants' abilities. Thereafter, the set undergoes an increase or decrease of one item, contingent on the participants' recognition responses, respectively correct or wrong.

Trial categorization as either 'correct' or 'error' is contingent upon the occurrence of an error during the flanker task. The authors observe that error trials resulted in a decrease in the level of accuracy on the recognition of the letter's sequence in comparison to correct trials, which is known as the ERIAM effect (Error Related Impairment of Active Working Memory). In Experiment 2, the ERIAM effect demonstrate a medium effect size ($t(52) = 3.29, p = .002$, Cohen's $d = 0.45$). However, no increase in response times on span recognition is observed, which would then more specifically signify a speed-accuracy trade-off adjustment (Donkin et al., 2014). In which case where the results are associated with a response times' decrease, a decrease in working memory recall's accuracy can be more simply explained by a decrease in the time taken by subjects to formulate and provide the correct response. Furthermore, the authors increased the inter-stimulus interval (from 200 to 2300 milliseconds in Experiment 1 and Experiment 2, respectively) and once again obtained an ERIAM effect with a longer interval. This finding demonstrates that the ERIAM effect is not contingent on the bottleneck passage of information postulated by some non-functional theories of PES (Jentzsch & Dudschig, 2009). In addition, the ERIAM effect is more pronounced in low span participants than in high span participants, thereby providing a link between WM capacities and error

monitoring abilities (Unsworth et al., 2022).

The study conducted by Wessel et al. (2022), while offering interesting results, can be subject to a few limitations, particularly on the methodological level. One limitation, due to post-hoc hypothesizing of a link between the ERIAM effect and WM capacities, is impossible experimental control of claim. In fact, over the course of five experiments in their study, Wessel and colleagues observe a negative correlation between ERIAM effect and participants' WM capacities ($r = -0.41, p < .001$). High span individuals are less impacted by motor error than are low span individuals. One experiment in the protocol proposes to increase the number of items in the memory span. Similar in all respects to the previous experiment, the authors no longer observe this significant correlation between the ERIAM effect and WM capacities with this cognitive load's increase ($r = -0.37, p = .071$). This observation needs further exploration and this work attempts to address it by systematically varying the number of items to be memorized.

The present study will observe the effects of an error on a digit span in working memory by implementing a complex span protocol. This type of protocol, for example reading span (Daneman & Carpenter, 1980), operation span (Turner & Engle, 1989) or symmetry span (Kane et al., 2004) requires participants to maintain items in memory while performing a second concurrent task. The maintenance of working memory and the selection of salient aspects of the stimuli within the same task are both called upon. The error on the conflict task will then allow for the categorization of trials according to their accuracy and for the estimation of the effect of the error on the maintenance of the working memory span.

The primary objective of this study is to replicate the ERIAM effect (Wessel et al., 2022) by implementing a modified version of Experiment 2, where the most significant ERIAM effect was observed. Trials are conducted in a laboratory setting to ensure the collection of the most precise data, unencumbered by the constraints of online runs related to connections and/or hardware. The objective is twofold: firstly, to provide new data to retest the finding of Wessel et al. (2022); and secondly, to attempt to verify Adaptive Orienting Theory's (Wessel, 2017) prediction of error-induced inhibition of ongoing cognitive processes during the task. The WM task is to be closed with digit span recall rather than recognition. This approach is underpinned by the observation that the recognition response to the WM task engages the same response modalities as the conflict task, both of which involve choosing between two possibilities: YES/NO for the similarity to the set of digits and Left/Right for the direction of the target arrow on the arrow task. The use of the numeric keypad keys to recall the digit span facilitates the dissociation of these response modalities and the observation of performance on the WM task by minimizing the confusion of response modalities (Pashler, 1990).

Furthermore, Wessel et al. (2022) demonstrated a significant negative correlation (in five out of seven experiments) between the ERIAM effect and subjects' average WMC, concluding that the ERIAM effect more strongly impacts subjects for whom the memory task is more demanding. Individuals with better WMC are those who are better able

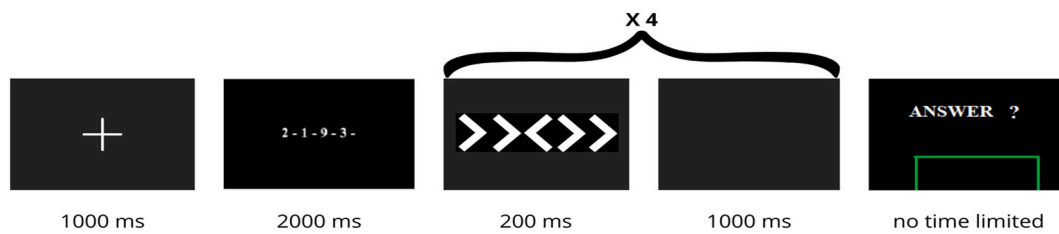


Fig. 1. Task procedure.

to regulate their attentional processes and to monitor the occurrence of error. The ERIAM effect appears to be less pronounced for individuals with higher working memory capacity (WMC) compared to those with lower WMC. However, a notable point that merits attention is the observation that the correlation between the ERIAM effect and WMC becomes marginal and non-significant when an increase in cognitive load is introduced, as evidenced by the transition from Experiment 5 to Experiment 6. This decline in the ERIAM/WMC correlation with increasing cognitive load is a subject that merits further investigation, particularly given that the two experiments that led to this difference in results involved two separate samples.

2. Presentation of hypotheses

Firstly, it is hypothesized that memory recall performance will deteriorate in the aftermath of a motor error (i.e., in the context of the flanker task). This initial hypothesis will be tested by replicating the ERIAM effect observed by Wessel et al. (2022) using a slightly modified task.

Secondly, during an increase in cognitive load, subjects classified as high span will have their attentional resources mobilized more extensively when recalling the span, and will have reduced ability to inhibit the effect of an error on their motor task. Subjects classified as low span will be less affected by the increase in cognitive load because their attentional resources will have been consumed at lower levels of load. We will test the ERIAM effect in these two groups and under these two conditions of cognitive load.

Thirdly, in the wake of our second hypothesis testing, we hereby propose an exploratory section (with analysis conducted post-hoc following hypothesis 2 testing) in an attempt to enhance our comprehension of the ERIAM effect. In this section, we shall undertake the testing of a more elementary construct (accuracy) within the group span and cognitive load conditions.

The fourth point to be considered concerns the ERIAM/WMC correlation, which will be observed in both groups and conditions of load. Kane and Engle (2000) demonstrated that high span individuals were more impacted by an increase in cognitive load than low span individuals. Thomassin et al. (2015) also identified a similar pattern in their study, finding that high span individuals' strategies are more efficient but also more sensitive to task's complexity. This effect was termed the 'Hard Fall Effect'. The present study hypothesizes that the correlation between ERIAM and WMC will become positive in the high span group as cognitive load increases. In contrast, it is expected that no such difference will be observed in the low span group, thus supporting our third hypothesis.

In conclusion, a secondary exploratory component is hereby proposed, encompassing a post-hoc analysis in accordance with hypothesis 3. In the absence of any consideration for cognitive load, the objective is to observe ERIAM/WMC relation for the whole group in alignment with the approach adopted by Wessel et al. (2022).

3. Methods

3.1. Participants/sample size justification

In order to justify the sample size, a proportion of the available information is utilized. The original study sought to attain 80 % power by collecting data from 53 participants. The information presented by Wessel et al. (2022) provided two possibilities. The first option is to utilize the effect size of their pilot study, while the second option is to employ the effect size obtained in their experiment 2, which is being replicated. Given that the effect size of the pilot study is more restrictive (0.39 versus 0.45), it is decided that this criterion would be selected in order to ensure the power necessary for the observation of the ERIAM effect. This estimated effect size also corresponds to the average expected effect size of 0.4 according to Brysbaert (2019) in psychology research. In order to ensure a satisfactory probability of replication, the a priori power is increased to 90 % (Erdfelder et al., 1996). Data on a sample of 63 participants are collected. The participants all gave their free and informed consent to undergo the experiment. The Research Ethics Committee of Toulouse approved the study by ruling on the ethical compliance of the present study (CER 2023-596). The study was pre-registered on OSF (Brochet et al., 2022). The task protocol, raw data, transformed database and analysis script are available to the public via the following link: https://osf.io/u6pyk/?view_only=c98aa9df704a46eaa39d8ef5c1eabe63.

Following the completion of the data processing stage, the final sample comprises 58 participants, with 68.25 % of these being female and 31.75 % male. The mean age of the sample is 25.84 years, with a standard deviation of 9.54 years.

3.2. Material/stimuli

The conflict task stimuli comprise a five-character string with symbols ">" and "<" (see Fig. 1). The width and height of these stimuli represent 10.8° and 3.3° of visual angle, respectively, and are centered on a 17-inch screen located 60 cm from the participants. The stimuli for the flanker task consist of a target and four flankers, distributed two by two on each side of the target (e.g. >><<>>). For the memory task, digits from 0 to 9 are presented in a Times New Roman font size 80. Each digit in a sequence is separated from the others by the '-' character to limit deliberate chunking (Gobet & Lane, 2012). The digits are constructed in a randomized manner, with potential repetition to extend the sequence beyond ten digits. All stimuli are presented on a white background against a black background. The task is programmed using Eprime 3.0 software (Psychology Software Tools, Pittsburgh, PA).

3.3. Procedure

The structure of the task is delineated in Fig. 1, with each trial exhibiting a consistent configuration. The initiation of each trial is marked by the appearance of a fixation cross, situated centrally on the screen for a duration of 1000 milliseconds. This is followed by the presentation of a sequence of digits at the same location on the screen for 2000 milliseconds. Subsequently, a neutral screen, characterized by a black background, is displayed for 200 milliseconds prior to the

commencement of the flanker task. The flanker task itself consists of a sequence of four stimuli for a total duration of 4800 milliseconds. Each stimulus is comprised of a target accompanied by four flankers, which are presented on the screen for 200 milliseconds. Thereafter, a black screen is presented for 1000 milliseconds, during which participants are permitted to respond. In contrast to the replicated study (100 milliseconds of advance for flankers, followed by the addition of the target for 100 milliseconds), there is no priming stage in this study. The four flankers all pointed in the same direction (left or right). The trial is considered to be congruent or incongruent depending on whether the central target pointed in the same direction as the flankers or the opposite direction. Participants are asked to respond to the orientation of the central target as quickly as possible by pressing the “A” or “P” keys for left and right, respectively. The four stimuli of the flanker task are followed by the memory recall task, during which participants are instructed to recall the digits presented at the beginning of the trial using the number pad and then validate with ENTER. There is no time limit defined on this recall task, and the next trial is presented only after participants have validated.

3.4. Training trials for span calibration

The experiment is composed of two sections. The first part consists of a training phase comprising 24 trials. During this phase, the length of the sequence of digits varies according to the participant's response accuracy. The objective of this phase is to calibrate the length of the sequence requested in the memory task (the span) to the subject's capacities. In order to calibrate the span during the training phase, the experiment protocol groups the trials 2 by 2. On these blocks of 2 trials, 3 possibilities arise depending on whether the participants had 0, 1 or 2 correct answers. When participants have zero correct answers, the protocol reduces the digits sequence by one unit. In instances where a single correct answer is attained, the protocol replicates the number of digits in the sequence presented. Finally, when participants have two correct answers, the protocol increases the sequence by one digit. The calibration phase commences with a sequence of four digits. Following the conclusion of the 24 training/calibration trials, the number of digits in the sequence attained by the participants is fixed. The subjects are not informed about the purpose of these calibration trials. The calibration portion consists exclusively of congruent flanker stimuli, thereby ensuring that:

- Participants could become familiar with structure of the trials and response modalities.
- Digit span recall is not overly impacted by motor errors in calibration phase.

In this training phase, participants receive feedback following the execution of each action. A green dot is awarded for a correct response, while a red dot indicates an error response. A response time that is more than the flanker task's time limit (200 ms) has a “Not fast enough” feedback message. Participants are not subject to a time limit for their responses in the digit span task. Notably, no participant achieves a memory digit span of less than four digits.

3.5. Completion of the task

The second part of the experiment consists of the task itself, which comprises 60 memory trials. Flanker stimuli are presented as congruent and incongruent (with a 50:50 % proportion) within each memory trial. In order to test the second hypothesis, an increase in digit span is measured by adding one digit to the sequence calibrated in the training phase on half of the trials. The total duration of the experiment is approximately 20 min.

Table 1
Summary of descriptive statistics on each task of the protocol.

	Task	Mean (SE) [95 % CI]
Accuracy on error trial	WORKING MEMORY	0.35 (0.23) [0.29, 0.41]
Accuracy on correct trial		0.44 (0.28) [0.37, 0.52]
RT on error trial		5.8 (1.7) [5.4, 6.3]
RT on correct trial	FLANKER	6 (1.6) [5.6, 6.4]
RT on congruent trial		0.44 (0.066) [0.43, 0.46]
RT on incongruent trial		0.51 (0.075) [0.49, 0.53]
Accuracy on congruent trial		0.95 (0.1) [0.92, 0.98]
Accuracy on incongruent trial		0.73 (0.15) [0.69, 0.77]

Note. For each task of the protocol, mean, standard error, and 95 % confidence interval are represented. For the Working Memory task, accuracy and response time (RT in second) are represented according to error or correct trial on Conflict Task. For flanker task, RT and accuracy are represented according to congruent or incongruent trials.

3.6. Data analysis

The behavioral data (i.e., accuracy and RT) from a portion of the task that does not include training/calibration trials is analyzed. Trials are eliminated in instances where participants exhibit an absence of response on the flanker task or memory task. Trials where response times on the flanker task are less than 200 ms are also deleted. The data from memory span responses are transformed into proportions of correct responses, as in the replicated study, to conduct the analyses. The analyses are conducted between subjects, with the data aggregated within participants, thus enabling the invocation of the central limit theorem.

In order to test our initial hypothesis, a two-sided paired-samples *t*-test at an alpha level of 0.029, corrected for a two-stage study (Pocock, 1977), is employed to compare the accuracy of digit span responses in error and correct conditions. These conditions are defined as follows: a trial is defined as an error if an error occurs in the interfering flanker task and as correct when no error is made on flanker task. The level of congruence is 50 % in both conditions in order to avoid confusing the ERIAM effect with a congruence effect.

In order to test the second hypothesis, which posits that an increase in cognitive load exerts a greater impact on the ERIAM effect in participants with a high span score, a linear mixed model is fitted (estimated using restricted maximum likelihood) to predict the ERIAM effect with cognitive load (CL) and high/low span group (formula: $ERIAM \sim CL * span + (1 | Subject)$). Groups are differentiated by number of digits in digit span (above and below the sample median). The cognitive load levels differ by the addition of one item in the digit span (for the increased load condition) compared to the moderate condition. The ERIAM effect is calculated as the difference in working memory recall accuracies between error and correct conditions on the conflict task.

In addition, the relation between the ERIAM effect and WMC is tested for the purpose of the study. WMC is defined as the overall response accuracy on memory task, excluding any consideration of error and correct conditions on the flanker task. Comparison is made between the high span and low span groups, as well as across both moderate and increased levels of cognitive load.

4. Results

An interim data analysis was conducted on this study (at N/2 in April 2023). In order to preserve the significance of the hypothesis tests, the Pocock correction (Pocock, 1977) is applied to the alpha thresholds of the primary hypotheses. Employing a two-stage design with two looks at the data, the Pocock correction indicates an alpha level of 0.029 for two-tailed tests. Consequently, this alpha level is applied consistently throughout the remainder of the study. The summary statistics are presented in Table 1.

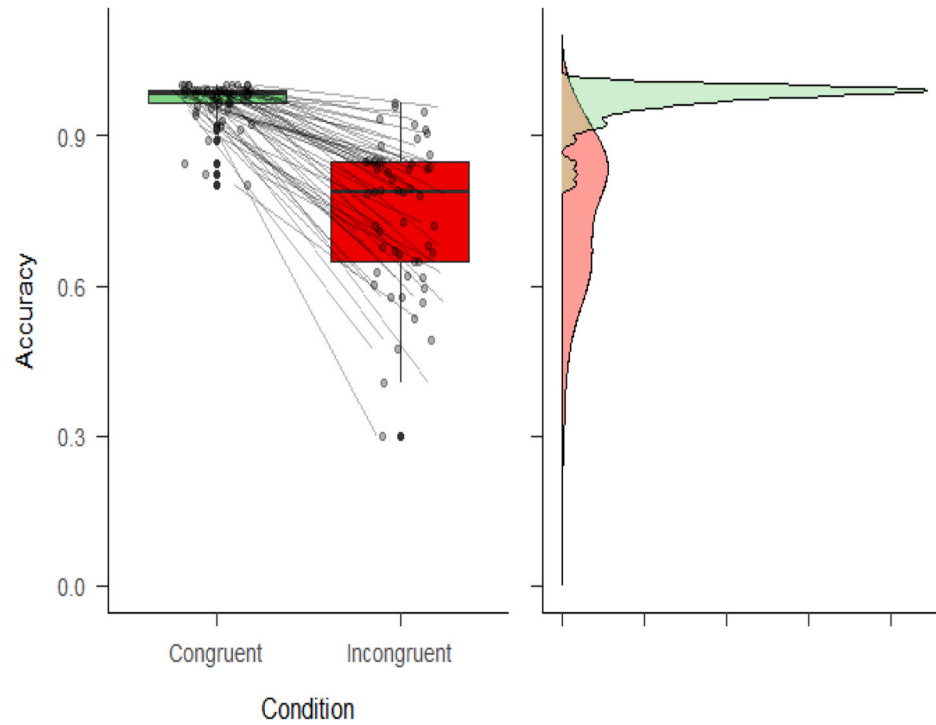
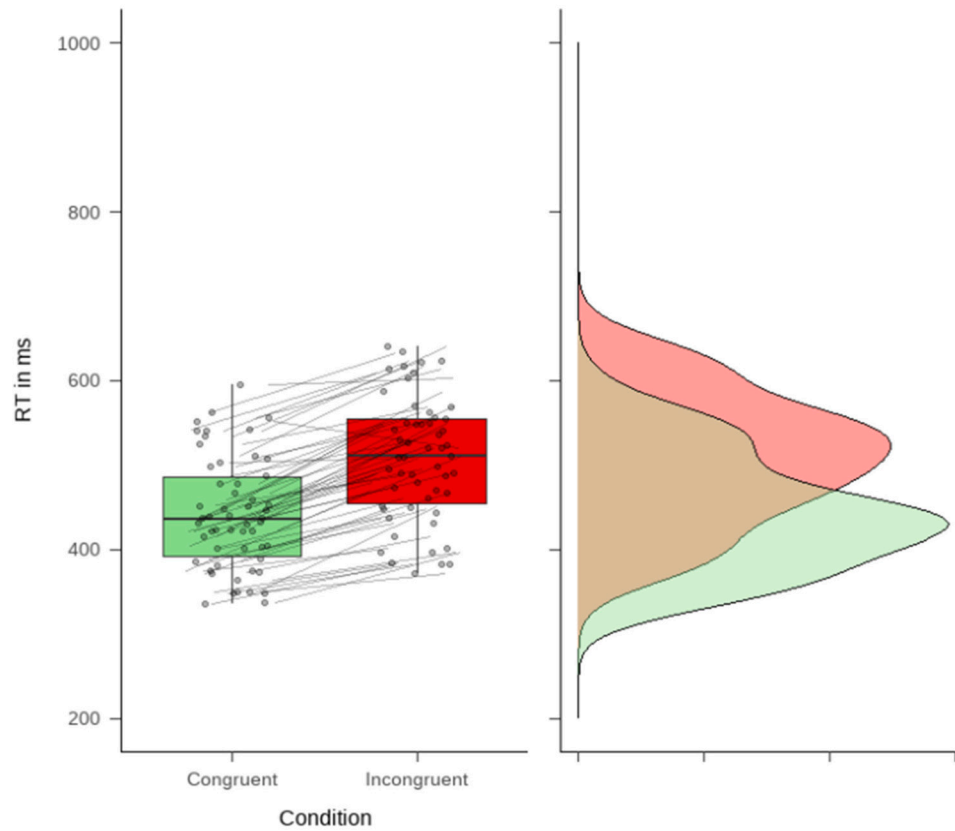
(a): Accuracy on conflict task**(b): Response time on conflict task**

Fig. 2. (a): Accuracy on conflict task.
 (b): Response time on conflict task.

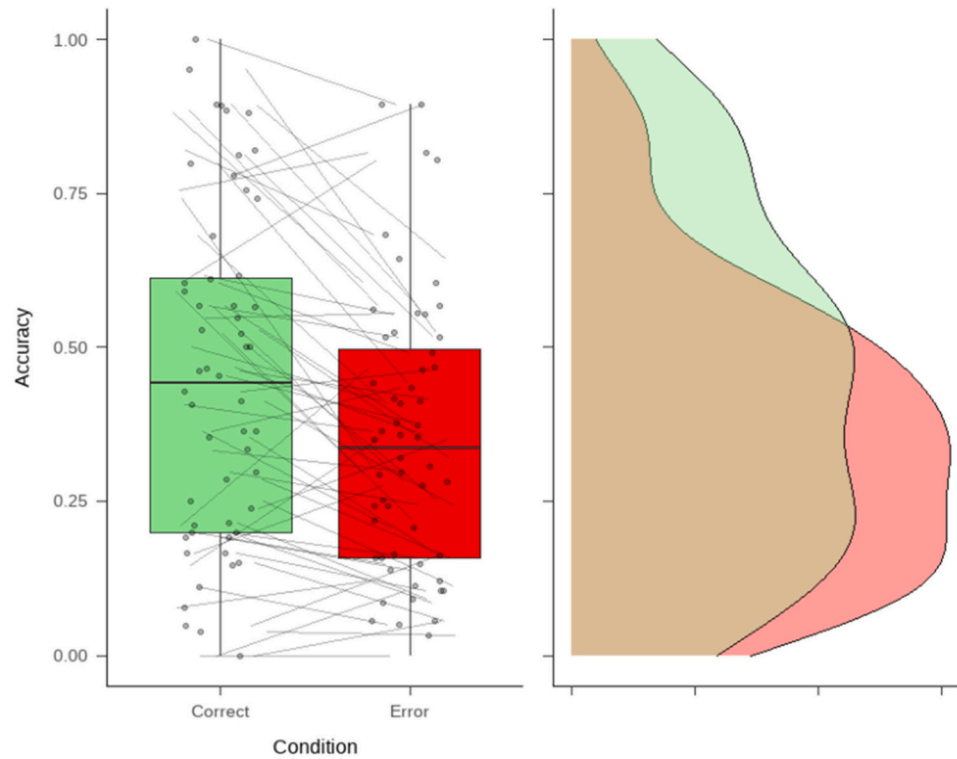
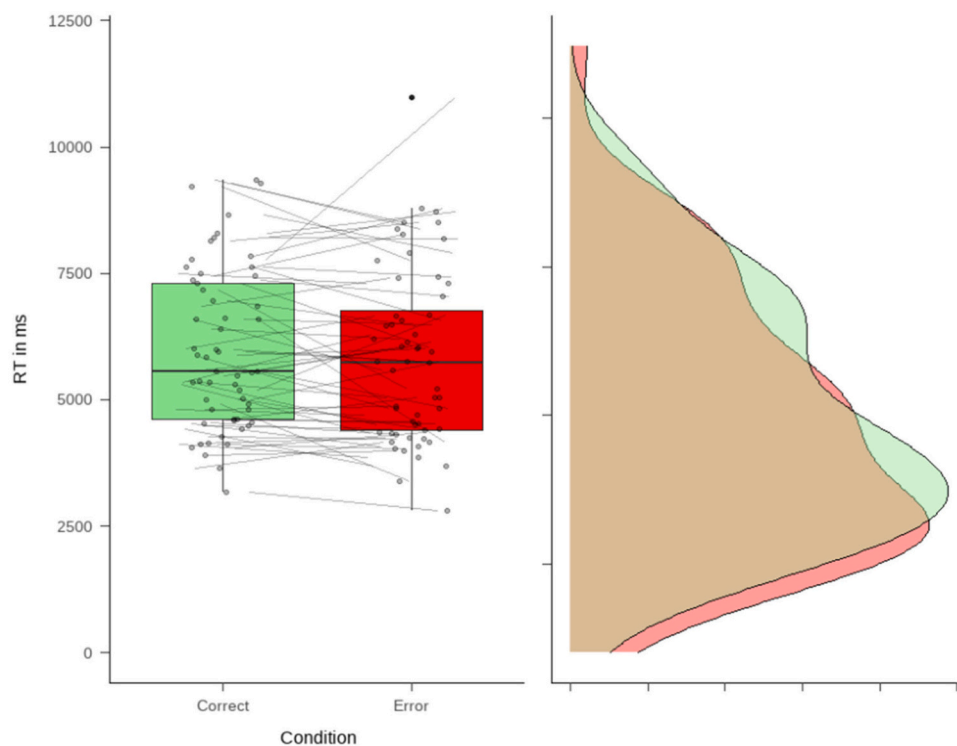
(a): Accuracy on Working Memory Task**(b): Response time on Working Memory Task**

Fig. 3. (a): Accuracy on Working Memory Task.
 (b): Response time on Working Memory Task.

4.1. Hypothesis 1 testing

4.1.1. Conflict task results

An illustration of the results observed on the conflict task is offered in

Fig. 2 (a & b), summary statistics is available in Table 1. Two participants were excluded from the analyses because of below-chance responses on the flanker task. Compared to congruent trials, incongruent trials are less accurate ($t(53) = 12.14, p < .001$, Cohen's $d = 1.79$, 95 %

Box 2

What about span, WMC and cognitive load?

Span is a distinctive feature of this study following the conclusion of the calibration phase, in which the quantity of digits presented is subject to variation, contingent on whether the response is accurate or erroneous. Span thus proposes to divide the sample into two groups (**high and low span**) according to the number of digits obtained. Participants with fewer than seven digits (the median) are designated as low span, while those with more than seven digits are classified as high span. Just remember that high spans retain more digits than low spans.

WMC is the **overall mean accuracy** on Working Memory task, as in Wessel et al. (2022). Counter-intuitively, low spans may have a better WMC than high spans! They have fewer numbers to remember, so the task may be easier for them.

Cognitive load is also associated with the quantity of digits. In the present study, the cognitive load is varied by introducing an additional number to the series presented for recall. This clearly relates the cognitive load to the **difficulty of the task**. It is imperative to acknowledge that the nomenclature employed (i.e., high and low span) does not reflect the inherent capacity of individuals in the working memory task (WMC). This capacity is measured in addition as the average response accuracy on the working memory task, without taking into account the number of digits reached by the participants. The number of digits in the span is thus employed in the creation of groups and the objective of varying the cognitive load (and consequently the complexity of the task). However, this does not serve to define the participants' working memory capacity. Nevertheless, the terms "high" and "low" span are retained, given that these are participants with a high or low number of digits in the span. Fig. 4 provides graphical support for these considerations, offering a visualization of WMC by cognitive load with a dotted red line showing the delimitation between the low and high span groups.

$CI = [0.19, 0.26]$, *point estimate* = 0.22) and have longer RTs than congruent trials ($t(53) = 15.16$, $p < .001$, Cohen's $d = 0.94$, 95 % $CI = [0.06, 0.07]$, *point estimate* = 0.07). This finding suggests that the manipulation employed in the conflict task is effective in replicating the conflict interference effects that have been extensively documented in the literature.

4.1.2. Memory task results

An illustration of the observed results is shown in Fig. 3(a & b), summary statistics is available in Table 1. The memory load of the participants consists of 4 to 10 digits ($M = 6.97$, $SD = 1.25$). The primary research question of this study, which aims to replicate the work of Wessel et al. (2022) and test the initial hypothesis, concerns the impact of an error on the conflict task on subsequent WM recall. The data reveal no substantial deviation from a normal distribution.

In this study, response accuracy of working memory recall in two distinct conditions is examined: those where an error is produced in the conflict task and those where no error is produced. The results reveals that trials which involved an error in the conflict task demonstrate significantly lower levels of accuracy in comparison to those trials which do not involve an error ($t(53) = 4.48$, $p < .001$, Cohen's $d = 0.61$, 95 % $CI = [0.05, 0.14]$, *point estimate* = 0.09). This effect is larger than that observed in Experiment 2 of Wessel et al. (2022), where the authors found a Cohen's d of 0.45. Consequently, response accuracy falls on average from 0.44 to 0.35, respectively, between the correct and error conditions (see Fig. 3a). This effect does not appear to be linked to an adjustment of the participants' response time/accuracy function (see Fig. 3b), as response times of two conditions do not differ significantly ($t(55) = 1.13$, $p = .265$, Cohen's $d = 0.08$, 95 % $CI = [-99.09, 352.88]$, *point estimate* = 126.89). This suggests that action errors committed during the delay period impair subsequent working memory accuracy on the digit span recall. This finding lends further support to our initial hypothesis and serves to replicate the ERIAM effect. Concerning this hypothesis, the a posteriori power analysis indicates a power of 98,49 %.

Before continuing with our presentation, a few points need to be made (Box 2).

4.2. Hypothesis 2 testing

The subsequent objective is to ascertain whether the ERIAM effect exerts a differential impact on high and low span groups within the two cognitive load conditions. To this end, a linear mixed model employing restricted maximum likelihood (REML) is utilized to evaluate the variation in ERIAM level in accordance with cognitive load and subject span

group. The formula employed for this purpose is as follows:

$$eriam \sim cognitive\ load * span + (1|subject)$$

Mixed models are a statistical tool that is employed to analyse data that exhibits both repeated measures within a subject and non-independence of data. The approach involves model comparison, whereby a null model is initially fitted using this formula:

$$eriam \sim 1 + (1|subject)$$

The intraclass correlation coefficient (ICC) is found to be 0.6, indicating a strong data dependency within the subject. The design effect demonstrates that the sample size is artificially increased by a factor of 2.59, with the random effect not being taken into consideration. Mixed model is appropriated in this case. A comparison of the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) is therefore made. Compared to the *null model*, first model with span as a predictor do not yield significant improvement of fit to the data. $AIC = 506.27$ for the *null model* and $AIC = 506.68$ for the model with predictors while $BIC = 516.2$ and 506.68 respectively for *null model* and model with predictors. *Bayes Factor* do not really favor any model ($Bf_{10} = 0.156$, $p = .098$). Changing predictor for cognitive load leads to same conclusion with an AIC of 505.44, this time favoring model with predictor but BIC of 518.67 and *Bayes Factor* ($Bf_{10} = 0.291$, $p = .017$) in line with the null model. Adding both predictors (cognitive load and span) to the model exhibits a significantly worse fit to the data compared to the null model ($AIC = 508.41$, $BIC = 519.91$). Once again, *Bayes Factor* favors *null model* ($Bf_{10} = 0.002$, $p = .024$). Consequently, the model with predictors is not analyzed, although the results shows a surprising trend towards a negative effect of increased cognitive load on the ERIAM effect ($\beta = -0.22$, 95 % $CI [-0.41, -0.04]$, $t(198) = -2.43$, $p = .016$) which can also be seen in Fig. 5. Consequently, according to the analyses performed, the second hypothesis of a more pronounced ERIAM effect in the high span group with increased cognitive load compared to the low span group is not supported by the data. Nevertheless, outside the scope of our hypothesis, we observe that the increase in cognitive load seems to have a negative impact on the ERIAM effect.

4.2.1. A posteriori power analysis

In order to shed light on these non-significant results and to enable future studies to adjust their protocols as effectively as possible, an a posteriori statistical power test was performed. The statistical power obtained for the test of each of the model's predictors has been tested. It should be noted that the interaction effect was not subjected to a power analysis, due to its complexity. For the effect of Cognitive Load predictor

Table 2

Model comparison with AIC and BIC.

Model	AIC	BIC
Null	14,288.70	14,303.64
Model 1	14,276.41	14,298.81
Model 2	13,390.16	13,420.02
Model 3	13,277.43	13,314.77
Full model	13,267.13	13,334.33

Table 3

Accuracy mean values in each condition tested in model.

Flanker trials	Cognitive load	Span	Accuracy
Correct	Moderate	Low span	0.74
		High span	0.42
	Increased	Low span	0.44
		High span	0.17
Error	Moderate	Low span	0.63
		High span	0.31
	Increased	Low span	0.32
		High span	0.11

on ERIAM, the statistical power obtained is 68.7 %. Furthermore, a visual analysis of the power curve (Fig. 6) suggests that a statistical power of 80 %, which is considered a reasonable threshold (Cohen, 1992), would be achieved with a sample of 70 participants.

For the effect of Span group predictor on ERIAM, the statistical power obtained is 38.3 %. Furthermore, a visual analysis of the power curve (Fig. 7) suggests that a statistical power of 80 % would be achieved with a sample of 100 participants.

4.2.2. Exploratory part 1 for in-depth understanding of the ERIAM effect

Going further in our understanding of the pattern beyond the data collected, analysis of accuracy on working memory task is carried as a more low-level construct than ERIAM. As accuracy is a binary variable at trial level, a logistic regression model using maximum likelihood (Laplace Approximation) is fitted to the data to assess for change in response accuracy on working memory task according to cognitive load, subject span group and error on conflict task.

The subsequent step involves model comparison, whereby the predictors are added individually and then the interactions for the full model are incorporated. The models that are subjected to this process are listed below:

- Null model: $accuracy \sim 1 + (1|subject)$
- Model 1: $accuracy \sim span + (1|subject)$
- Model 2: $accuracy \sim cognitive\ load + span + (1|subject)$
- Model 3: $accuracy \sim cognitive\ load + span + error + (1|subject)$
- Full model: $accuracy \sim cognitive\ load * span * error + (1|subject)$

Table 2 shows the AIC and BIC values for the different models.

Considering AIC and BIC, it is evident that two models are worthy of particular consideration: the Model 3 and the Full model. These two models incorporate all the predictors. However, it is challenging to differentiate between these models based on these criteria, as AIC and BIC do not favor the same model. Nevertheless, it can be concluded that

the model without interactions is more parsimonious and therefore to be preferred. Model 3 is the most suitable model for data interpretation, as it exhibits the highest degree of compatibility with the data set:

$$accuracy \sim cognitive\ load + span + error + (1|subject)$$

Therefore, a generalized linear mixed model (GLMM) with a binomial family and logit link function is fitted on data to examine the effects of span group, cognitive load, and error on accuracy, while accounting for random variability across subjects. The model incorporated span (low vs. high span), cognitive load (moderate vs. increased), and error (correct vs. error on flanker task) as fixed effects, with subject designated as a random intercept. Table 3 shows the accuracy mean values for each condition.

The random effect of subject had a variance of 1.52 (SD = 1.23), indicating substantial variability across participants. For the fixed effects, span significantly predicted accuracy ($B = 1.37$, $SE = 0.34$, $z = 4.08$, $p < .001$), indicating that low span participants had significantly higher accuracy compared to high span participants. Cognitive load also had a significant effect ($B = -1.30$, $SE = 0.05$, $z = -28.19$, $p < .001$) with increased load leading to lower accuracy. Additionally, the effect of error was found to be significant ($B = -0.51$, $SE = 0.05$, $z = -10.71$, $p < .001$) indicating that trials with flanker errors are associated with lower accuracy thus showing an alternative method for attesting to the ERIAM effect.

The collective findings imply that group span, cognitive load, and errors play a substantial role in predicting task accuracy. A notable finding is that low span participants outperformed high span participants in terms of accuracy, which is contrary to the initial expectations. Fig. 8 provides graphical representations of the model and a possible insight to interpret this: Response accuracy can be seen decreasing from left to right, suggesting that the grouping done by span could be interpreted as another type of variation in cognitive load, with a more cognitively challenging task in high spans. Finally, the impact of increased cognitive load and errors on performance is in line with the expected negative outcomes.

4.3. Hypothesis 3

4.3.1. Let's start exploring (exploratory part 2)

As stated in the study's preregistration (Brochet et al., 2022), the relation between ERIAM and WMC is examined, as outlined in Wessel et al. (2022). In this analysis, participants with an ERIAM of 0 are excluded from the database. This is statistically improbable, and the result can be attributed to two possibilities: a participant with perfect or zero accuracy scores in all conditions. It is hypothesized that this would then be indicative of a participant having failed to adhere to the task instructions, either by ignoring the flanker task entirely or by responding randomly to the task. For the whole group, a Spearman's rank-order correlation is conducted to assess the relationship between ERIAM and WMC. The Spearman test is chosen in order to take into account a possible non-linear relationship between WMC and ERIAM. The analysis yields a moderate positive correlation, $\rho = 0.38$, 95 % CI = [0.09, 0.6], $p = .004$. This finding suggests a positive relationship between WMC and ERIAM, whereby an increase in WMC is associated with a concomitant increase in ERIAM, contrary to the findings of the replicated study

Table 4

ERIAM effect, WMC and Spearman rank order correlation between ERIAM and WMC in high and low span groups according to cognitive load (CL).

Group	Cognitive Load	ERIAM	SEM	p value	ERIAM/WMC	p value	WMC	SEM	95 % IC	N
lowspan	Moderate	0.15	0.05	0.003	-0.01	0.948	0.64	0.05	[0.54, 0.74]	23
	Increased	0.09	0.04	0.068	0.53	0.012	0.4	0.06	[0.28, 0.52]	22
highspan	Moderate	0.07	0.03	0.042	0.24	0.187	0.38	0.04	[0.3, 0.46]	32
	Increased	0.07	0.02	0.014	-0.11	0.618	0.25	0.03	[0.19, 0.31]	24

It is noteworthy that the ERIAM effect is only significant in lowspan group in the moderate cognitive load condition and in highspan group in the increased load condition. This particular pattern will be covered in the discussion section.

(Wessel et al., 2022).

Table 4 presents the ERIAM, WMC and Spearman rank order correlation ERIAM/WMC values across both participant groups at distinct levels of cognitive load. It is noteworthy that alpha threshold for statistical significance was set at 0.029, following Pocock's correction. The findings reveal that the ERIAM effect appears to be significant only in the high span/increased load and low span/moderate load conditions, while the correlation ERIAM/WMC is significant in the low span group in the increased load condition.

4.3.2. Hypothesis testing

The objective of this hypothesis is to ascertain whether the correlation between ERIAM and WMC become positive in high spans when cognitive load increased and remains constant in low spans. As demonstrated in Table 4, this appears not to be the case. Spearman rank order correlation tests are performed on the data to test this hypothesis, the statistical power of each test ($1-\beta$) are also reported. For the low span group in the moderate load condition, Spearman test is not significant ($\rho(22) = -0.01, p = .948, 1-\beta = 0.03$). For the low span group in the increased load condition, the test is significant ($\rho(21) = 0.53, p = .012, 1-\beta = 0.67$). Contrary to what has been assumed, in low span the link between ERIAM and WMC becomes positive if cognitive load is increased. This pattern is not found in high-span individuals. Indeed, with these, Spearman test yields $\rho(31) = 0.24, p = .187, 1-\beta = 0.2$ in the moderate load condition and $\rho(23) = -0.11, p = .618, 1-\beta = 0.05$ in the increased load condition. The third hypothesis posits an increase in the strength of the positive link between ERIAM and WMC in the high span group when cognitive load is elevated. This hypothesis is verified through the implementation of a Steiger test (Steiger, 1980), giving a non-significant result ($z = 1.12, p = .262$). However, third hypothesis cannot be substantiated by the data collected. However, it can be seen that the statistical power of these tests is generally very low.

4.3.3. Going back to exploration (exploratory part 3)

While the overall positive correlation between the ERIAM effect and WMC is surprisingly different from that observed by Wessel et al. (2022) and significant, the latest results are not conclusive. However, these results suggest that there may be a more complex, non-linear link between ERIAM and WMC. As illustrated in Fig. 9, the WMC - ERIAM scatterplot is presented alongside quartile regression lines ($\tau = 0.25, 0.5, 0.75$) and the overall regression curve, the latter of which has been determined using the Locally Estimated Scatterplot Smoothing method.

ERIAM and WMC do not differ from a normal distribution, with respective Shapiro Wilk test values of $W = 0.98, p = .701$ for ERIAM and $W = 0.96, p = .107$ for WMC. The distribution of ERIAM appears to be symmetrical overall, while WMC demonstrates a marginal propensity towards positive skewness ($skewness = 0.29$). While an overall the link between ERIAM and WMC appears positive, an inflection seems to be present at the highest WMC levels.

4.4. Supplementary control analyses

To address the potential confound that the correlation between WMC and ERIAM may be affected by the covariance structure between correct error trials, Wessel et al. (2022) conducted a non-parametric statistical test with a bootstrapping method. However, the results of this test are not presented in their article. In order to check this confound, it is proposed to reproduce these analyses with the data from the present study. The methodology is reproduced identically to the study by Wessel et al. (2022), as followed:

We simulated subject-level WM accuracy for Mixed-correct [correct] and Mixed-error [error] conditions using two groups of randomly generated numbers, with the constraints that (1) the group size was set to the number of subjects in this experiment; (2) the means and covariance structure of empirical data were used in the simulation;

and (3) the correlation coefficient of between the two groups of simulated data differed from the cross-subject correlation coefficient between Mixed-correct [correct] and Mix-error [error] accuracy by less than 0.01. To simulate the predicted correlation, Pearson's r was calculated between the mean across the two groups (simulating overall WM performance) and their difference (simulating ERIAM). This procedure was repeated for 10,000 times to estimate a null distribution of correlation coefficient between overall WM accuracy and ERIAM, while controlling for mean difference and covariance structure of WM accuracy between Mixed-correct [correct] and Mixed-error [error] conditions. P-values derived from the null distribution were one-tailed, as the null distribution was not necessarily centered at 0 or symmetric.

This analysis was conducted three times: for the whole group, the low span and the high span. Fig. 10 illustrates distribution of simulated correlation coefficients and empirical correlations (using Pearson's r).

As can be seen, the correlations obtained empirically all differ significantly from the bootstrap distributions ($r = 0.36, p < .001$ for the whole group, $r = 0.28, p < .001$ for the low span group and $r = 0.31, p < .001$ for the high span group). On this point, therefore, the results of hypothesis 3 should be treated with great caution. It is important, therefore, that further research should add to our understanding of this ERIAM/WMC relationship.

5. Discussion

In order to test the first hypothesis, the effect of a motor error on working memory maintenance (ERIAM) was examined through a replication and extension of the second experiment from Wessel et al. (2022). This study successfully replicates ERIAM effect at the group level. Memory recall was significantly impaired following errors on the flanker task. Wessel et al. (2022) suggest that in experiments employing an arrow flanker task, the ERIAM effect may be facilitated by the similarity of the response modalities across the two task components, both of which involve dichotomous responses. In the present study, however, the recall phase was operationalized differently – requiring participants to recall a digit rather than to recognize one. This methodological variation demonstrates that the ERIAM effect persists even when the response modalities differ and do not rely on the same response keys. Furthermore, the effect remains observable despite the absence of time pressure during the working memory task, suggesting that the ERIAM effect endures over longer delay intervals following an error, likely indicative of a degradation in memory trace.

To understand these results, it is necessary to refer back to Adaptive Orienting Theory (AOT; Wessel, 2017). AOT predicts that errors are resulting from an impairment of cognitive processes in progress. Memory tasks involve encoding, storage (with rehearsal for WM) and retrieval of information. The impact of errors on the retrieval process can likely be ruled out (but further studies would be needed to rule this out definitively), as the recall task was not time-limited, allowing participants to perform equally across error and correct conditions. However, it remains unclear which of the remaining two processes – encoding or storage – is affected by the error. Assuming that encoding occurs prior to the onset of the conflict task and the occurrence of the motor error, it is plausible that errors interfere with rehearsal processes in working memory, by diverting limited attentional resources used for its maintenance (Barrouillet & Camos, 2020).

As Oberauer (2019) observes, among the theoretical accounts conceptualizing WM as dependant on attentional resource, the notion of a shared resource for both storage and processing has garnered the most theoretical support. Within this framework, and in line with functional theories of PES, error detection engages deployment of cognitive control to be monitored (Botvinick et al., 2001). As this process unfolds, cognitive resources are redirected towards error monitoring, thereby diverting some of the attentional resources allocated to information

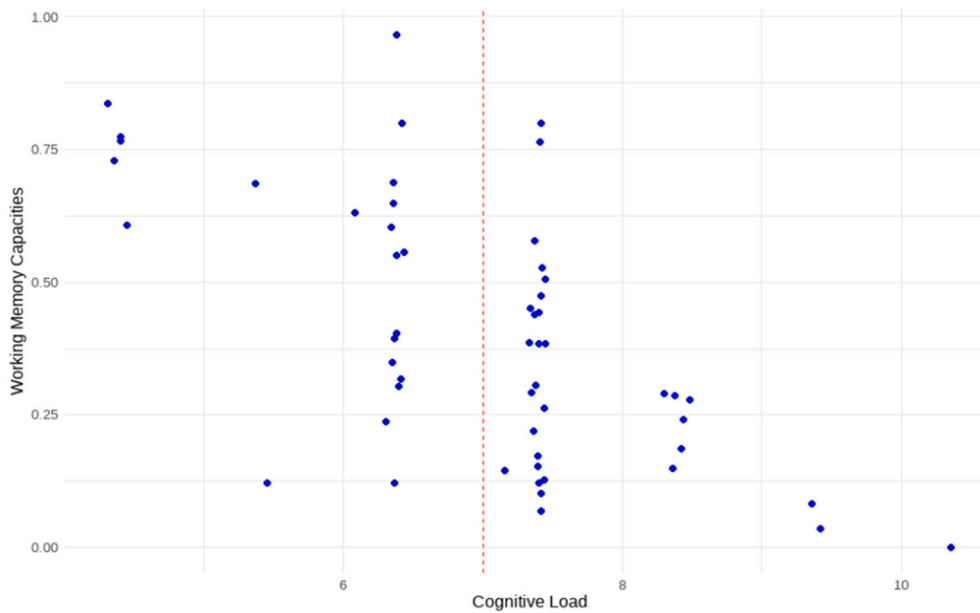


Fig. 4. WMC/Cognitive Load scatterplot with separation line for group span.

maintenance in working memory. It is therefore essential to disentangle these processes by developing alternative protocols specifically designed to study these constructs.

In relation to the second hypothesis, which concerns the differential impact of increased cognitive load on high- and low-span groups, a hierarchical model using REML is implemented to predict ERIAM, taking into account the non-independence of data. However, findings indicate that neither cognitive load nor group span are effective predictors of the ERIAM effect in the sample. This is probably because ERIAM is a highly transformed summary measure derived from data (δ between mean correct and mean error accuracy). Obviously, an alternative explanation is the lack of statistical power, with an a posteriori power of 68.7 % for *cognitive load* and 38.3 % for *span* predictors. Alternatively, this may be a consequence of the mixed-effects modelling approach, in which case all variance would fall into the random factor. In this regard, the conclusions drawn in the present study remain insufficient, and further investigation is required.

In the subsequent phase of the study, an exploratory analysis is conducted with the aim of predicting accuracy based on span group, cognitive load and error occurrence on the conflict task. Accuracy is a lower-level construct than ERIAM, and revisiting this level serves to provide a more simplified understanding of data. The model including these predictors demonstrates a significantly better fit to the data compared to the null model. This model reveals that all main effects are significant: committing an error on the conflict task (ERIAM effect), increasing cognitive load and belonging to the high span group each have a significant negative effect on accuracy.

To further elaborate on these main effects, it is first evident that the error factor exerts a significant negative effect on accuracy. This finding aligns with expectations, given the previously identified ERIAM effect during the testing of Hypothesis 1. Notably, in this analysis, the fixed effect of error is identified while accounting for random effect. In addition, the findings indicate that cognitive load exerts a detrimental effect on accuracy across all participants. Once more, this outcome is anticipated, as increasing the digit span by one, although individually calibrated, further increases task complexity. Then, participants in the high span group exhibit lower level of accuracy in comparison to those in the low span group. This unexpected result is likely attributable to the experimental design, which defines span groups based on the maximum digit span achieved during the training phase. It is plausible that high span participants encounter trials that are more challenging, thereby

resulting in significantly lower performance relative to low span participants.

The third hypothesis of the study posited that there would be a more pronounced positive association between ERIAM and WMC in the high span group when cognitive load is increased. However, this hypothesis is not supported by the data. The correlation between ERIAM and WMC, after controlling for group span and cognitive load, only reached significance in the low span group and the high cognitive load condition. These results are still likely due to the analysis involving complex constructs (ERIAM and WMC being derived from highly transformed data) and requiring greater statistical power (as shown by the statistical power) to reach significance.

Notwithstanding, the exploratory section shows that the general link between WMC and ERIAM appears to be positive overall. This relationship does not appear to be strictly linear and constant as a function of WMC, but tends to reverse at higher WMC values (while acknowledging the exploratory nature of this part). It is possible that in Wessel et al. (2022), where the precision of responses is higher than in the present study (0.8 and 0.76 in Wessel et al. (2022) compared to 0.44 and 0.35 in present study for correct and error condition, respectively), the negative link between WMC and ERIAM is therefore part of the regression curve (Fig. 9) where a downward slope is observed. It would therefore be particularly interesting for future research to investigate this link more closely. Therefore, it appears that the Wessel et al. (2022) proposal that “ERIAM most affected subjects for whom the WM aspect of the task was more challenging” may warrant a more nuanced interpretation.

High span participants demonstrated less impaired performance on digit span recall following an error on the flanker task (i.e., lower general ERIAM level, see Fig. 5 and Table 4). However, their overall accuracy (WMC) was lower than that of the low span group, which is likely attributable to the greater levels of task complexity encountered by this group. Conversely, low span participants exhibited a higher propensity for ERIAM alongside higher level of WMC. This finding underscores the observed positive association between ERIAM and WMC in our sample, as higher WMC is attained by low span participants, likely due to the lower task complexity in this group. Wessel et al. (2022) posit that “The likely explanation for this finding [positive correlation between ERIAM and WMC] is that subjects with a lower overall WM accuracy maintain less stable WM representations, which are then more susceptible to the impairing effects of errors.” The current study, which shows lower overall WM accuracy among high span individuals -likely due to a ceiling effect due

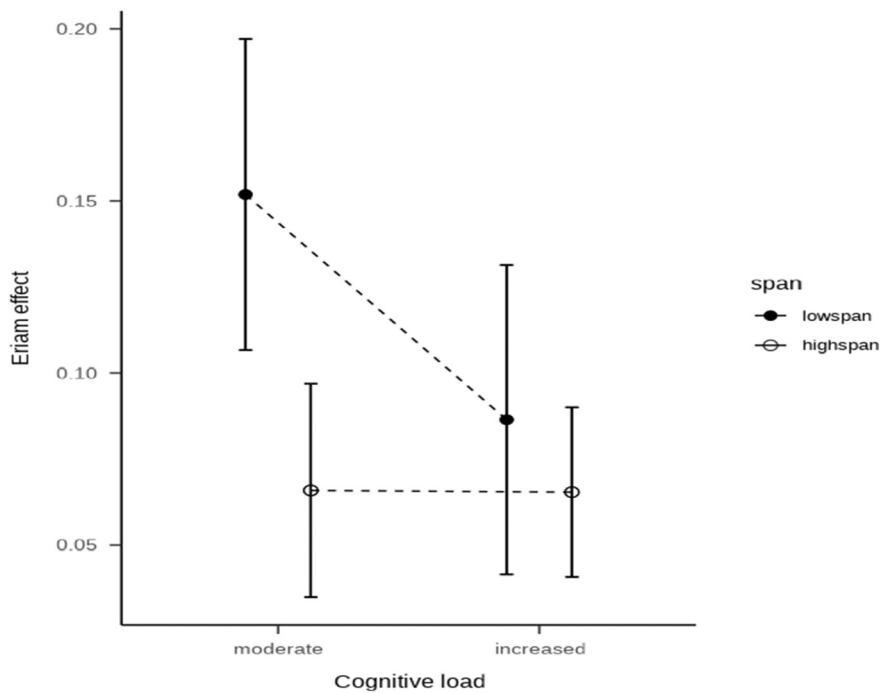


Fig. 5. ERIAM effect according to cognitive load and group.
Note: None of the results yield significance.

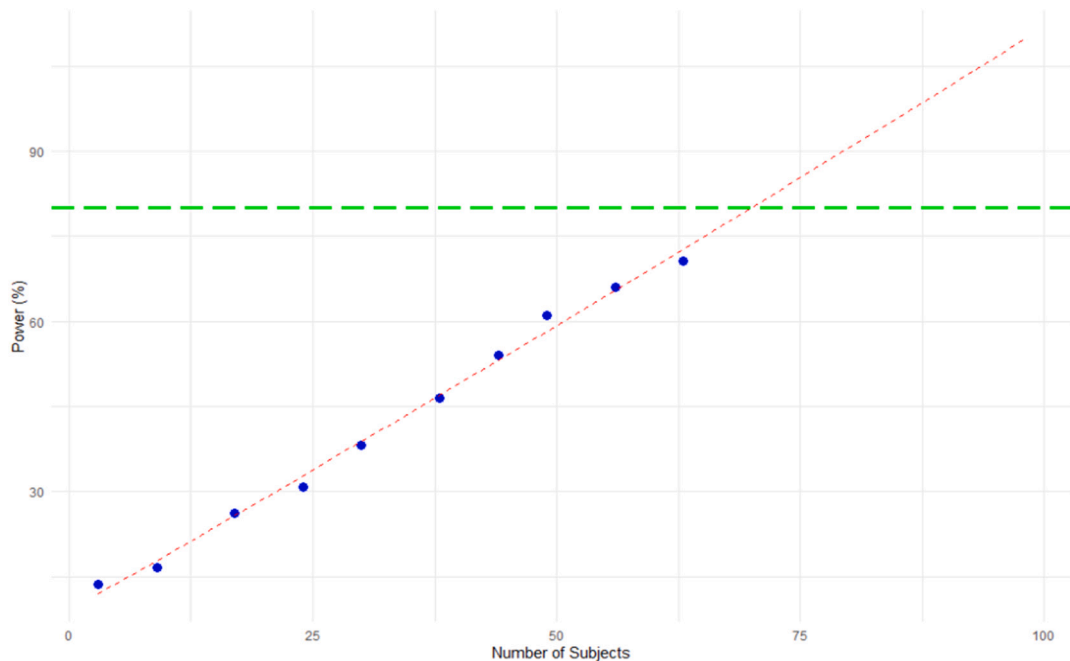


Fig. 6. Power curve with extrapolation for Cognitive Load.

to task complexity- together with positive association between ERIAM and WMC suggests that the complexity of the memory task may constitute an important moderating factor of the ERIAM effect.

According to Lavie (2010), “The ability to focus attention [...] deteriorates under conditions of high load on cognitive control processes such as working memory.” In the present study, the findings appear to support this aforementioned theoretical prediction, as accuracy levels on the working memory task seem to decrease as cognitive load increases (Table 3, Fig. 4 and Fig. 8). Lavie’s load theory encompasses the effects of both perceptual and cognitive load; while high perceptual load reduces

the processing of irrelevant stimuli, high cognitive load increases it. Returning to the PEAs functional and non-functional explanations, it is widely accepted that an error constitutes an irrelevant stimulus to the task at hand. Errors divert attention and may or may not engage further processing, depending on the theoretical account, to regain maximum performance levels. Among the factors that may determine whether error monitoring is engaged, and whether the ERIAM effect is observed, cognitive and perceptual load levels represent plausible candidates.

The ERIAM effect is only significant in the low span group in the moderate cognitive load condition and in the high-span group under the

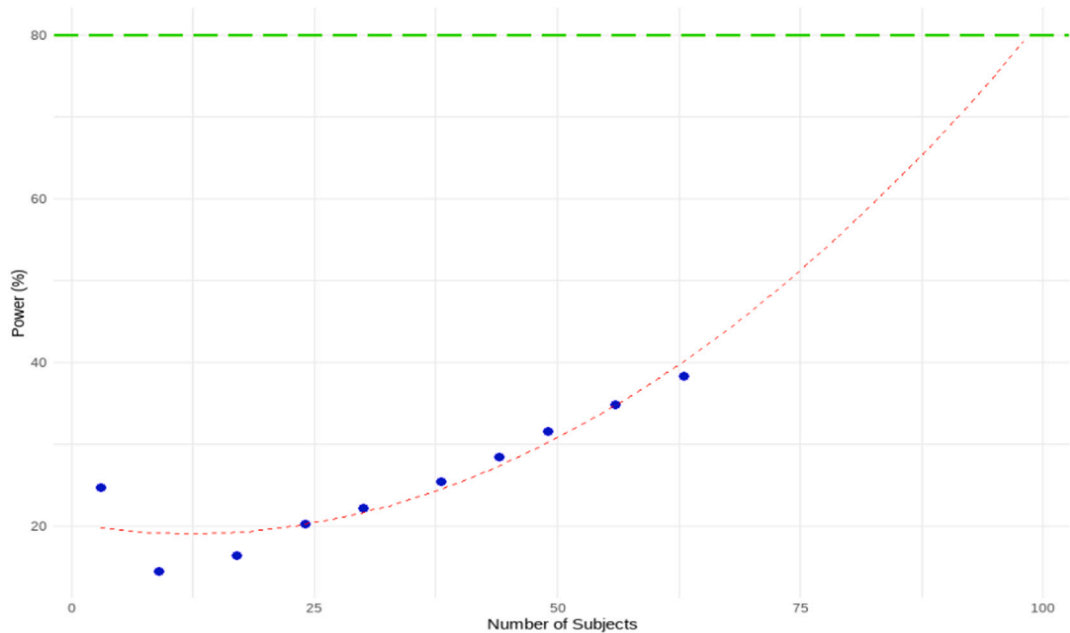


Fig. 7. Power curve with extrapolation for Span group.

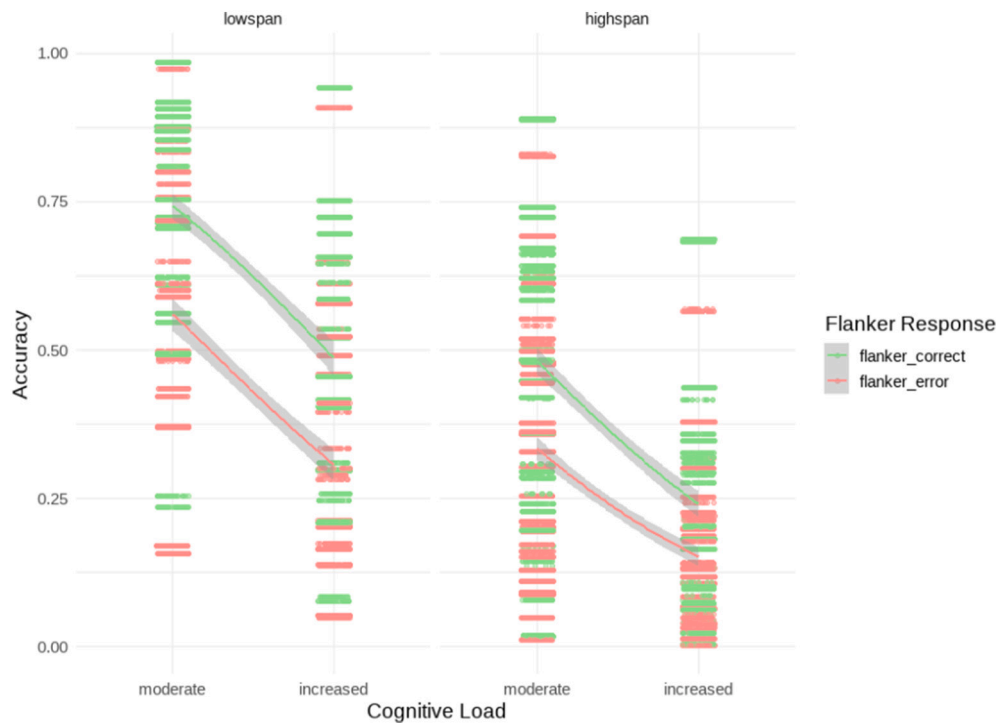


Fig. 8. Accuracy on memory task according to span, error and cognitive load as predicted by the logistic regression model.

increased load condition (see Table 4). Referring to Lavie's load theory, it can be presumed that for low span participants, the increased cognitive load condition exceeds their cognitive capacities, thereby impairing their ability to sustain focus attention. Low span participants likely disengage cognitive control processes (including error monitoring) from the task, which may explain why ERIAM effect is not significant in this condition. For this group, it is plausible that errors under increased cognitive load are filtered out at the perceptual level, failing to trigger additional processing and, consequently, not leading to ERIAM. Conversely, an opposite pattern is observed in high span participants. High spans under increased cognitive load condition likely engage

cognitive control more fully in response to the task demands, but due to being overburdened by task complexity, they exhibit a stronger ERIAM effect. The interference effect of error is maximized in this group by the increase in cognitive load, as predicted by Lavie's theory. Nevertheless, this group demonstrates a greater capacity to manage error occurrence under conditions of moderate cognitive load.

6. Conclusion

To conclude, the present study provides significant insights into the relationship between motor errors and Working Memory maintenance

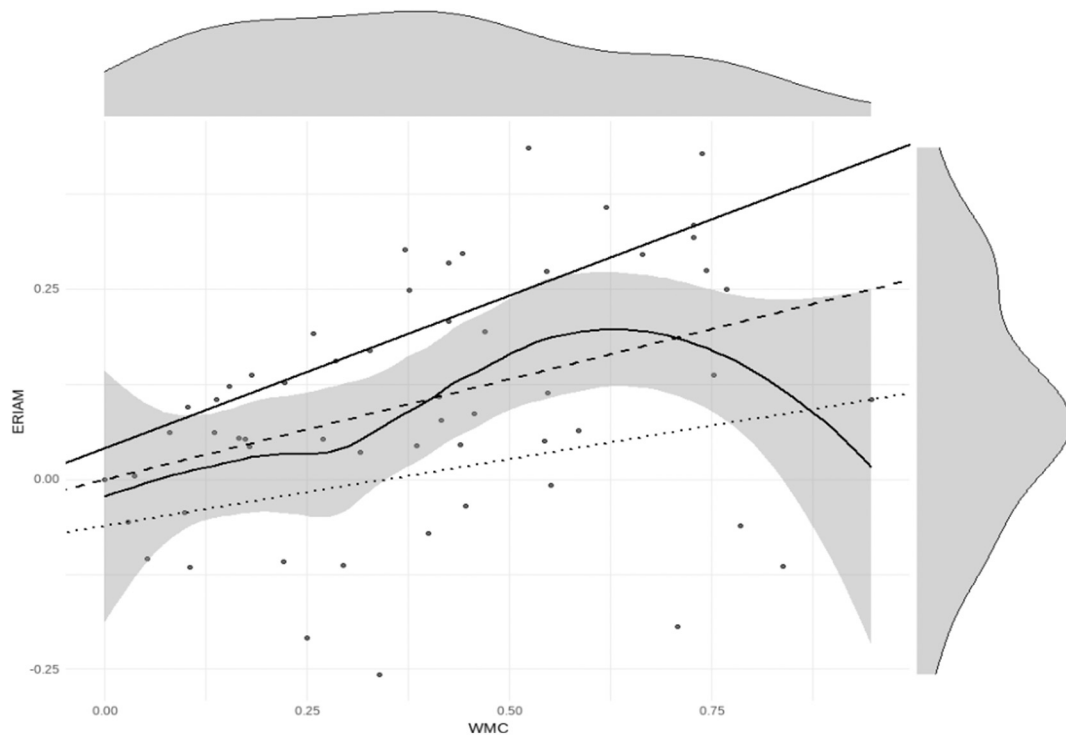


Fig. 9. ERIAM – WMC scatterplot with quartile regression lines and overall regression curve using loess method. Note: Dotted line, dashed line and solid line respectively represent 0.25, 0.5 and 0.75 quartile.

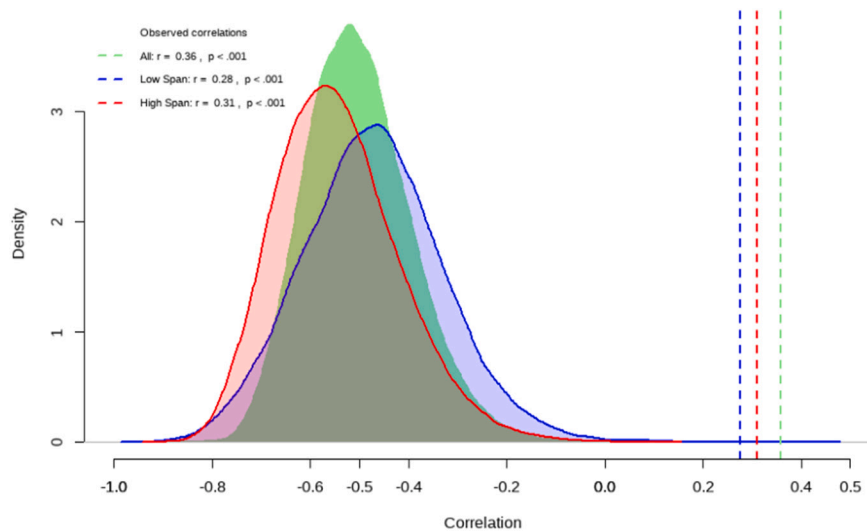


Fig. 10. Simulated and empirical correlations between WMC and ERIAM.

(ERIAM), as well as the influence of cognitive load and individual differences in working memory capacity. Firstly, the study successfully replicates previous findings demonstrating that motor errors impair memory recall, extending this effect to tasks with different response modalities, indicating a broader applicability of ERIAM across task variations. The results obtained in this study are consistent with the Adaptive Orienting Theory (Wessel, 2017), which suggests that the process of detecting errors diverts attentional resources from memory maintenance, particularly rehearsal in working memory. However, the precise impact on encoding versus storage remains inconclusive and requires further investigation.

As Wessel et al. (2022) assert, “[...] errors consistently produce non-selective impairments of WM maintenance across all tasks when the WM

aspect of the task is sufficiently difficult.” The present study corroborates these statements, while broadening their scope. Indeed, it has been observed that if the task becomes excessively complex, the ERIAM effect tends to diminish (in low span), and conversely, if the task is not complex enough, the ERIAM effect also tends to diminish (in high span). Consequently, subsequent studies may investigate the potential of ERIAM as a behavioral indicator of cognitive engagement in task.

The findings of this study contradict the results of previous research regarding the relationship between ERIAM and Working Memory Capacity (WMC). An overall positive relationship was observed in this study. This discrepancy may be attributed to differences in task complexity and individual cognitive strategies. High span individuals exhibited lower ERIAM levels but also lower WMC, possibly indicating a

ceiling effect due to task complexity for this group.

The present study contributes to the understanding of how motor errors impact working memory maintenance, shedding light on the moderating roles of cognitive load and individual differences. Future research should further elucidate the underlying mechanisms of ERIAM and its implications for cognitive control processes. For instance, in order to further examine the effect of a motor error on working memory processes, it is proposed that neurophysiological aspects be investigated, including a potential correlation with error-related negativity (Falkenstein et al., 1991).

CRedit authorship contribution statement

Emilien Brochet: Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pierre-Vincent Paubel:** Software. **Céline Lemerrier:** Supervision, Writing – original draft.

Declaration of competing interest

The authors declare there is no conflict of interest in this study.

Data availability

Data will be made available on request.

References

- Armitage, S. G. (1946). An analysis of certain psychological tests used for the evaluation of brain injury. *Psychological Monographs*, 60(1), i–48. <https://doi.org/10.1037/h0093567>
- Barrouillet, P., & Camos, V. (2020). *The time-based resource-sharing model of working memory* (pp. 85–115). Oxford University Press. <https://doi.org/10.1093/oso/9780198842286.003.0004>
- Botvinick, M. M., Braver, T. S., Barch, D. M., Carter, C. S., & Cohen, J. D. (2001). Conflict monitoring and cognitive control. *Psychological Review*, 108(3), 624–652. <https://doi.org/10.1037/0033-295x.108.3.624>
- Botvinick, M. M., Cohen, J. D., & Carter, C. S. (2004). Conflict monitoring and anterior cingulate cortex: An update. *Trends in Cognitive Sciences*, 8(12), 539–546. <https://doi.org/10.1016/j.tics.2004.10.003>
- Broadbent, D. E. (1958). *The selective nature of learning* (pp. 244–267). Elsevier. <https://doi.org/10.1016/b978-1-4832-0079-8.50012-8>
- Brochet, E., Lemerrier, C., & Paubel, P.-V. (2022). Error related impairment of active working memory (wessel et al., 2022) replication study. *Open Science Framework*. <https://doi.org/10.17605/OSF.IO/FYN4C>
- Brysbaert, M. (2019). How many participants do we have to include in properly powered experiments? A tutorial of power analysis with reference tables. *Journal of Cognition*, 2(1), 16.
- Buzzell, G. A., Beatty, P. J., Paquette, N. A., Roberts, D. M., & McDonald, C. G. (2017). Error-induced blindness: Error detection leads to impaired sensory processing and lower accuracy at short response stimulus intervals. *The Journal of Neuroscience*, 37(11), 2895–2903. <https://doi.org/10.1523/JNEUROSCI.1202-16.2017>
- Carp, J., & Compton, R. J. (2009). Alpha power is influenced by performance errors. *Psychophysiology*, 46(2), 336–343. <https://doi.org/10.1111/j.1469-8986.2008.00773.x>
- Christoff, K., Irving, Z. C., Fox, K. C. R., Spreng, R. N., & Andrews-Hanna, J. R. (2016). Mind-wandering as spontaneous thought: A dynamic framework. *Nature Reviews Neuroscience*, 17(11), 718–731. <https://doi.org/10.1038/nrn.2016.113>
- Cohen, J. (1992). Statistical power analysis. *Current Directions in Psychological Science*, 1(3), 98–101.
- Daneman, M., & Carpenter, P. A. (1980). Individual differences in working memory and reading. *Journal of Verbal Learning and Verbal Behavior*, 19(4), 450–466. [https://doi.org/10.1016/s0022-5371\(80\)90312-6](https://doi.org/10.1016/s0022-5371(80)90312-6)
- Danielmeier, C., Eichele, T., Forstmann, B. U., Tittgemeyer, M., & Ullsperger, M. (2011). Posterior medial frontal cortex activity predicts post-error adaptations in task-related visual and motor areas. *The Journal of Neuroscience*, 31(5), 1780–1789. <https://doi.org/10.1523/jneurosci.4299-10.2011>
- Danielmeier, C., & Ullsperger, M. (2011). Post-error adjustments. *Frontiers in Psychology*, 2. <https://doi.org/10.3389/fpsyg.2011.00233>
- Debenet, S., Ullsperger, M., Siegel, M., Fiehler, K., Cramon, D. Y. v., & Engel, A. K. (2005). Trial-by-trial coupling of concurrent electroencephalogram and functional magnetic resonance imaging identifies the dynamics of performance monitoring. *The Journal of Neuroscience*, 25(50), 11730–11737. <https://doi.org/10.1523/jneurosci.3286-05.2005>
- Derrfuss, J., Danielmeier, C., Klein, T. A., Fischer, A. G., & Ullsperger, M. (2021). Unbiased post-error slowing in interference tasks: A confound and a simple solution. *Behavior Research Methods*, 54(3), 1416–1427. <https://doi.org/10.3758/s13428-021-01673-8>
- Deutsch, J. A., & Deutsch, D. (1963). Attention: Some theoretical considerations. *Psychological Review*, 70(1), 80–90. <https://doi.org/10.1037/h0039515>
- Donkin, C., Little, D. R., & Hout, J. W. (2014). Assessing the speed-accuracy trade-off effect on the capacity of information processing. *Journal of Experimental Psychology: Human Perception and Performance*, 40(3), 1183–1202. <https://doi.org/10.1037/a0035947>
- Dudschig, C., & Jentzsch, I. (2009). Speeding before and slowing after errors: Is it all just strategy? *Brain Research*, 1296, 56–62. <https://doi.org/10.1016/j.brainres.2009.08.009>
- Dutilh, G., Ravenzwaaij, D. v., Nieuwenhuis, S., Maas, H. L. J. v. d., Forstmann, B. U., & Wagenmakers, E.-J. (2012). How to measure post-error slowing: A confound and a simple solution. *Journal of Mathematical Psychology*, 56(3), 208–216. <https://doi.org/10.1016/j.jmp.2012.04.001>
- Dutilh, G., Vandekerckhove, J., Forstmann, B. U., Keuleers, E., Brysbaert, M., & Wagenmakers, E.-J. (2011). *Testing theories of post-error slowing*. American Psychological Association (APA). <https://doi.org/10.1037/e520602012-241>
- Egeth, H., & Kahneman, D. (1975). Attention and effort. *The American Journal of Psychology*, 88(2), 339. <https://doi.org/10.2307/1421603>
- Egner, T. (2007). Congruency sequence effects and cognitive control. *Cognitive, Affective, & Behavioral Neuroscience*, 7(4), 380–390. <https://doi.org/10.3758/cabn.7.4.380>
- Engle, R. W., & Kane, M. J. (2003). *Executive attention, working memory capacity, and a two-factor theory of cognitive control* (pp. 145–199). Elsevier. [https://doi.org/10.1016/s0079-7421\(03\)44005-x](https://doi.org/10.1016/s0079-7421(03)44005-x)
- Erdfelder, E., Faul, F., & Buchner, A. (1996). GPOWER: A general power analysis program. *Behavior Research Methods, Instruments, & Computers*, 28(1), 1–11. <https://doi.org/10.3758/bf03203630>
- Eriksen, B. A., & Eriksen, C. W. (1974). Effects of noise letters upon the identification of a target letter in a nonsearch task. *Perception & Psychophysics*, 16(1), 143–149. <https://doi.org/10.3758/bf03203267>
- Falkenstein, M., Hohnsbein, J., Hoormann, J., & Blanke, L. (1991). Effects of crossmodal divided attention on late ERP components. II. Error processing in choice reaction tasks. *Electroencephalography and Clinical Neurophysiology*, 78(6), 447–455.
- Fiehler, K., Ullsperger, M., & Von Cramon, D. Y. (2005). Electrophysiological correlates of error correction. *Psychophysiology*, 42(1), 72–82. <https://doi.org/10.1111/j.1469-8986.2005.00265.x>
- Fievez, F., Derosiere, G., Verbruggen, F., & Duque, J. (2022). Post-error slowing reflects the joint impact of adaptive and maladaptive processes during decision making. *Frontiers in Human Neuroscience*, 16. <https://doi.org/10.3389/fnhum.2022.864590>
- Forster, S., & Lavie, N. (2007). High perceptual load makes everybody equal. *Psychological Science*, 18(5), 377–381. <https://doi.org/10.1111/j.1467-9280.2007.01908.x>
- Friedman, N. P., & Robbins, T. W. (2022). The role of prefrontal cortex in cognitive control and executive function. *Neuropsychopharmacology*, 47(1), 72–89. <https://doi.org/10.1038/s41386-021-01132-0>
- Gehring, W. J., & Fencsik, D. E. (2001). Functions of the medial frontal cortex in the processing of conflict and errors. *The Journal of Neuroscience*, 21(23), 9430–9437. <https://doi.org/10.1523/jneurosci.21-23-09430.2001>
- Gobet, F., & Lane, P. C. R. (2012). *Chunking mechanisms and learning* (pp. 541–544). Springer US. https://doi.org/10.1007/978-1-4419-1428-6_1731
- Gratton, G., Cooper, P., Fabiani, M., Carter, C. S., & Karayanidis, F. (2017). Dynamics of cognitive control: Theoretical bases, paradigms, and a view for the future. *Psychophysiology*, 55(3). <https://doi.org/10.1111/psyp.13016>
- James, W. (1890). *The principles of psychology* (vol. i). Henry Holt; Co. <https://doi.org/10.1037/10538-000>
- Jentzsch, I., & Dudschig, C. (2009). Short article: Why do we slow down after an error? Mechanisms underlying the effects of posterror slowing. *Quarterly Journal of Experimental Psychology*, 62(2), 209–218. <https://doi.org/10.1080/17470210802240655>
- Kane, M. J., & Engle, R. W. (2000). Working-memory capacity, proactive interference, and divided attention: Limits on long-term memory retrieval. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 26(2), 336–358. <https://doi.org/10.1037/0278-7393.26.2.336>
- Kane, M. J., & Engle, R. W. (2002). The role of prefrontal cortex in working-memory capacity, executive attention, and general fluid intelligence: An individual-differences perspective. *Psychonomic Bulletin & Review*, 9(4), 637–671. <https://doi.org/10.3758/bf03196323>
- Kane, M. J., Hambrick, D. Z., Tuholski, S. W., Wilhelm, O., Payne, T. W., & Engle, R. W. (2004). The generality of working memory capacity: A latent-variable approach to verbal and visuospatial memory span and reasoning. *Journal of Experimental Psychology: General*, 133(2), 189–217. <https://doi.org/10.1037/0096-3445.133.2.189>
- King, J. A., Korb, F. M., von Cramon, D. Y., & Ullsperger, M. (2010). Post-error behavioral adjustments are facilitated by activation and suppression of task-relevant and task-irrelevant information processing. *The Journal of Neuroscience*, 30(38), 12759–12769. <https://doi.org/10.1523/jneurosci.3274-10.2010>
- Laming, D. (1979). Choice reaction performance following an error. *Acta Psychologica*, 43(3), 199–224. [https://doi.org/10.1016/0001-6918\(79\)90026-x](https://doi.org/10.1016/0001-6918(79)90026-x)
- Lavie, N. (2010). Attention, distraction, and cognitive control under load. *Current Directions in Psychological Science*, 19(3), 143–148. <https://doi.org/10.1177/0963721410370295>
- Lavie, N., & Dalton, P. (2014). In A. C. (K.). Nobre, & S. Kastner (Eds.), *Load theory of attention and cognitive control*. Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780199675111.013.003>

- Lavie, N., & Tsai, Y. (1994). Perceptual load as a major determinant of the locus of selection in visual attention. *Perception & Psychophysics*, 56(2), 183–197. <https://doi.org/10.3758/bf03213897>
- Maier, M. E., Yeung, N., & Steinhauser, M. (2011). Error-related brain activity and adjustments of selective attention following errors. *NeuroImage*, 56(4), 2339–2347. <https://doi.org/10.1016/j.neuroimage.2011.03.083>
- Marco-Pallarés, J., Camara, E., Münte, T. F., & Rodríguez-Fornells, A. (2008). Neural mechanisms underlying adaptive actions after slips. *Journal of Cognitive Neuroscience*, 20(9), 1595–1610. <https://doi.org/10.1162/jocn.2008.20117>
- McIsaac, T. L., Lamberg, E. M., & Muratori, L. M. (2015). Building a framework for a dual task taxonomy. *BioMed Research International*, 2015, 1–10. <https://doi.org/10.1155/2015/591475>
- Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex “frontal lobe” tasks: A latent variable analysis. *Cognitive Psychology*, 41(1), 49–100. <https://doi.org/10.1006/cogp.1999.0734>
- Niedermeyer, E. (1994). Clinical neuropsychology of attention. *Electroencephalography and Clinical Neurophysiology*, 91(3), 236. [https://doi.org/10.1016/0013-4694\(94\)90083-3](https://doi.org/10.1016/0013-4694(94)90083-3)
- Notebaert, W., Houtman, F., Opstal, F. V., Gevers, W., Fias, W., & Verguts, T. (2009). Post-error slowing: An orienting account. *Cognition*, 111(2), 275–279. <https://doi.org/10.1016/j.cognition.2009.02.002>
- Oberauer, K. (2019). Working memory and attention - A conceptual analysis and review. *Journal of Cognition*, 2(1), 36. <https://doi.org/10.5334/joc.58>
- Pashler, H. (1990). Do response modality effects support multiprocessor models of divided attention? *Journal of Experimental Psychology: Human Perception and Performance*, 16(4), 826–842. <https://doi.org/10.1037/0096-1523.16.4.826>
- Pfister, R., & Foerster, A. (2021). How to measure post-error slowing: The case of pre-error speeding. *Behavior Research Methods*, 54(1), 435–443. <https://doi.org/10.3758/s13428-021-01631-4>
- Pocock, S. J. (1977). Group sequential methods in the design and analysis of clinical trials. *Biometrika*, 64(2), 191–199. <https://doi.org/10.1093/biomet/64.2.191>
- Posner, M. I. (1980). Orienting of attention. *Quarterly Journal of Experimental Psychology*, 32(1), 3–25. <https://doi.org/10.1080/00335558008248231>
- Rabbitt, P. M. (1966). Errors and error correction in choice-response tasks. *Journal of Experimental Psychology*, 71(2), 264–272. <https://doi.org/10.1037/h0022853>
- Rasmussen, J. (1983). Skills, rules, and knowledge; signals, signs, and symbols, and other distinctions in human performance models. *IEEE Transactions on Systems, Man, and Cybernetics, SMC-13*, (3), 257–266. <https://doi.org/10.1109/tsmc.1983.6313160>
- Ratcliff, R., & McKoon, G. (2008). The diffusion decision model: Theory and data for two-choice decision tasks. *Neural Computation*, 20(4), 873–922. <https://doi.org/10.1162/neco.2008.12-06-420>
- Reason, J. (1990). *Human error*. <https://doi.org/10.1017/cbo9781139062367>
- Regev, S., & Meiran, N. (2014). Post-error slowing is influenced by cognitive control demand. *Acta Psychologica*, 152, 10–18. <https://doi.org/10.1016/j.actpsy.2014.07.006>
- Schroder, H. S., Nickels, S., Cardenas, E., Breiger, M., Perlo, S., & Pizzagalli, D. A. (2020). Optimizing assessments of post-error slowing: A neurobehavioral investigation of a flanker task. *Psychophysiology*, 57(2), Article e13473. <https://doi.org/10.1111/psyp.13473>
- Shalgi, S., Barkan, I., & Deouell, L. Y. (2009). On the positive side of error processing: Error-awareness positivity revisited. *European Journal of Neuroscience*, 29(7), 1522–1532. <https://doi.org/10.1111/j.1460-9568.2009.06690.x>
- Steiger, J. H. (1980). Tests for comparing elements of a correlation matrix. *Psychological Bulletin*, 87(2), 245.
- Steinhauser, M., Ernst, B., & Ibal, K. W. (2017). Isolating component processes of posterror slowing with the psychological refractory period paradigm. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 43(4), 653–659. <https://doi.org/10.1037/xlm0000329>
- Thomassin, N., Gonthier, C., Guerraz, M., & Roulin, J.-L. (2015). The hard fall effect. *Experimental Psychology*, 62(2), 89–97. <https://doi.org/10.1027/1618-3169/a000276>
- Treisman, A. M. (1964). Selective attention in man. *British Medical Bulletin*, 20(1), 12–16. <https://doi.org/10.1093/oxfordjournals.bmb.a070274>
- Turner, M. L., & Engle, R. W. (1989). Is working memory capacity task dependent? *Journal of Memory and Language*, 28(2), 127–154. [https://doi.org/10.1016/0749-596x\(89\)90040-5](https://doi.org/10.1016/0749-596x(89)90040-5)
- Unsworth, N., Miller, A. L., & Robison, M. K. (2022). The influence of working memory capacity and lapses of attention for variation in error monitoring. *Cognitive, Affective, & Behavioral Neuroscience*, 22(3), 450–466. <https://doi.org/10.3758/s13415-022-01003-1>
- von Bastian, C. C., Blais, C., Brewer, G. A., Gyurkovics, M., Hedge, C., Katalamala, P., ... Wiemers, E. A. (2020). Advancing the understanding of individual differences in attentional control: Theoretical, methodological, and analytical considerations. <https://doi.org/10.31234/osf.io/x3b9k>
- Wessel, J. R. (2017). An adaptive orienting theory of error processing. *Psychophysiology*, 55(3), Article e13041. <https://doi.org/10.1111/psyp.13041>
- Wessel, J. R., Jiang, J., & Stolley, J. J. (2022). Action errors impair active working memory maintenance. *Journal of Experimental Psychology: General*, 151(6), 1325.
- Wickens, C. D. (2008). Multiple resources and mental workload. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 50(3), 449–455. <https://doi.org/10.1518/001872008x288394>